Sharif University of Technology

Department of Electrical Engineering

Course:

ASIC/FPGA

Report:

ASIC/FPGA Project - Group 8

Weekly Report IV

Submitted by:

Ali Behrad
401101321
Mohammadhossein Sabzalian
402101906
Sina Hashemi
402102668
Erik Hartouni
401102778

Instructor:

Dr. Baharvand

Department of Electrical Engineering

Term: Spring 2026

Project Management & Final Integration

Week 4 followed the guideline's Integration & Reporting phase. With the gate-level netlist timing-clean and functionally equivalent to RTL as of Week 3, the group closed out the project by re-validating that equivalence under full SDF back-annotation, packaging the deliverables for system hand-off, and finalizing the cross-group compatibility work with Group 7 that had been opened during the Week 2–3 testbench exchange. As Week 4 Integration Lead, this report was compiled by Ali Behrad (role D), with Mohammadhossein Sabzalian (role B) compiling the structural verification data and post-synthesis simulation analysis, Erik Hartouni (role C) finalizing the synthesis deliverables directory, and Sina Hashemi (role A) closing out the FSM timing correction found during cross-group testing.

Others:

- **Repository update:** The final deliverables package – RTL sources, testbench, synthesis scripts/constraints, the structural netlist and SDF, the Innovus setup script, and all Genus/Xcelium logs – was consolidated under a single hand-off tree, matching the Section 6.2 checklist below.
- **Scope of this week:** No new functional test cases were added; Week 4 work was gate-level verification closure (SDF back-annotation of the actual netlist, rather than only the timing reports read in Week 3), integration-facing documentation, and the FSM correction driven by Group 7's cross-testbench run.

Week 4 Plan vs. Actual

Planned Activity (Weekly Schedule)	Responsible	Status
Cross-group integration with Group 7	A, B, C, D	**Done** (Sec.)
Run top-level simulations with neighboring modules	A, B, C, D	**Done** – POR handshake and OTP-array timing verified (Sec.)
Prepare final report	C, D	**Done**
Prepare code package for system integration	A, B	**Done** (Sec.)
Final review, sign-off	A, B, C, D	**Done**
Integration Lead (Week 4) submits final package	D	**Done**

Table 1: Week 4 deliverables against the schedule defined in the project guideline.

Gate-Level Simulation & Back-Annotation Validation

GLS Compilation and SDF Back-Annotation Setup

Post-synthesis gate-level simulation was re-executed using the Cadence `xrun` engine, this time exercising the full timing-annotated flow end to end rather than the netlist/timing reports alone:

Listing 1: GLS Invocation with `POSTSYN_SIM` Defined

```
xrun -gui -ieee1364 -sv -disable_sem2009 -access +rwc -top
    tb_otp_controller \
+incdir+../rtl_sources +define+POSTSYN_SIM \
../rtl_sources/otp_sim_rom.v ../rtl_sources/tb_otp_controller.v
```

```
xrun -gui -ieee1364 -sv -disable_sem2009 -access +rwc -top tb_otp_controller +incdir+../rtl_sources
../rtl_sources/crc16_ccitt.v ../rtl_sources/otp_controller.v ../rtl_sources/otp_sim_rom.v
../rtl_sources/tb_otp_controller.v
```

Figure 1: Gate-level `xrun` invocation loading the `POSTSYN_SIM` testbench path.

Passing `+define+POSTSYN_SIM` switches the compile-time guard in `tb_otp_controller.v` (established in Week 3, Sec. **??**) to load the TSMC 180 nm gate models (`tsmc18.v`) and the Genus-mapped structural netlist in place of the behavioral `otp_controller.v`, and to back-annotate worst-case delays from `otp_controller_postsyn.sdf` onto `tb_otp_controller.u_dut` under **MAXIMUM** delay mode.

SDF Annotation Warnings Analysis

Elaboration (`xmelab`) flagged **476 warnings, 0 errors** while back-annotating the SDF:

```
xmelab: *W,SDFNET: Unable to annotate to non-existent timing check (RECREM (posedge RN) (posedge CK) (117) (0))
of instance tb_otp_controller.u_dut.ren_d_reg of module DFFRHQX1...

xmelab: *W,SDFNET: Unable to annotate to non-existent timing check (RECREM (posedge RN) (posedge CK) (117) (0))
of instance tb_otp_controller.u_dut.\aging_base_reg[0]  of module DFFRHQX1...
```

Figure 2: Representative `xmelab` SDFNET warnings on the asynchronous reset recovery/removal checks.

Warning Root Cause Every one of the 476 warnings is of the same `RECREM (posedge RN)(posedge CK)` form, flagged on the asynchronous reset pin (`RN`) of a DFFRHQX1/`SDFFRHQX1` sequential cell (e.g. `ren_d_reg`, `\aging_base_reg[0]`). Because `rst_n` was not constrained as an ideal clock net during synthesis, Genus did not compute recovery/removal timing checks for the reset path and left them uncalculated in the exported `.sdf`, so `xmelab` reports the corresponding timing check as *non-existent* rather than mis-annotated. This is a **library/constraint-scope** artifact, not a structural mismatch between the netlist and the SDF – confirmed by the **0 errors** count and, more importantly, by the fact that these are recovery/removal checks on an asynchronous control pin that is only exercised once, at POR, and is otherwise held stable for the remainder of the 256-cycle read sequence.

SDF Coverage Statistics

Despite the reset-path warnings, annotation of the timing paths that actually govern functional correctness was complete:

```
SDF statistics:
  No. of Pathdelays = 4979     Annotated = 100.00% (4979/4979)
  No. of Tchecks    = 2984     Annotated = 84.05%  (2508/2984)
            Total(T)   Annotated(A)   Percentage(A/T)
  Path Delays    4979       4979          100.00%
     $width     1428       1428          100.00%
   $setuphold    1556       1080           69.41%
```

Figure 3: SDF statistics: path-delay, `$width`, and `$setuphold` annotation coverage.

SDF Coverage Result **100.00%** (4,979/4,979) of gate-to-gate **path delays** and **100.00%** (1,428/1,428) of pulse-**width** checks were annotated. Only **$setuphold** checks were partial, at **69.41%** (1,080/1,556) – consistent with Section : the unannotated 476 checks are exactly the reset-pin recovery/removal arcs the library left uncalculated, and 1,556 − 1,080 = 476 confirms this is the same warning population, not a separate coverage gap. With every physical propagation delay and width check fully back-annotated, the simulation reflects realistic worst-case gate timing for the signals that determine the design's cycle-by-cycle behavior.

Post-Synthesis Simulation Run Results

The full 39-check regression (Weeks 2–3 baseline) was re-run against the back-annotated structural netlist:

```
====================================================
 SUMMARY: 39 PASSED, 0 FAILED (of 39 checks)
====================================================
 RESULT: ALL TESTS PASSED
Simulation complete via $finish(1) at time 17238606705 PS + 0
```

Figure 4: Final gate-level regression result: 39/39 checks passed, simulation completed via `$finish`.

Final GLS Regression **39 of 39 checks passed, 0 failed**, concluding with **RESULT: ALL TESTS PASSED**. The run terminated via `$finish(1)` at **17,238,606,705 ps** (≈17.24 ms), matching the RTL reference completion time to the picosecond. Because the data path delay through the mapped combinational logic (2.727 ns, Week 3) is nearly four orders of magnitude smaller than the 7.629 μs clock period, no metastable sampling or corrupted FSM transition was possible under back-annotated timing, and the physical netlist reproduced the read sequence, CRC-16 verification, and priority-ordered patch overrides identically to the behavioral RTL model.

Cross-Group Collaboration & FSM Timing Correction

Parallel Group Collaboration (G7 & G8)

Following the redundancy-verification requirement of the guideline's Section 5 (modules assigned to two groups), Group 8 continued its collaboration with the parallel OTP Controller group, Group 7, begun in Week 2–3: exchanging self-checking testbenches, comparing synthesis constraint strategies, and cross-simulating each group's RTL against the other's test suite.

FSM Timing Discrepancy & Resolution

The Issue: Under Group 7's stricter address-assertion checks, our original FSM updated the address counter (`addr`) and transitioned states with a one-clock-cycle offset between the two updates. This sequential mismatch caused the controller to remain one extra cycle in the active read states, so instead of reading exactly 256 bits (addresses 0–255), the design attempted a **257th, out-of-bounds read** – a discrepancy invisible to our own testbench but caught immediately by Group 7's assertions.

The Correction: Sina Hashemi restructured the FSM next-state logic and the address-counter increment path so that the state transition and the address update are driven from the same combinational look-ahead and commit on the identical clock edge, eliminating the one-cycle offset and restricting the read phase to exactly 256 cycles.

Cross-Verification Result After the correction, the updated RTL was re-run against both our own 39-check regression and Group 7's exchanged testbench. All timing and data-alignment assertions from both test suites passed, confirming the OTP Controller is compatible with the parallel group's independent implementation and closing the redundancy-verification requirement for this module.

Integration Notes for System Hand-off

To support drop-in integration with the neighboring analog and frequency modules, the following port-level interface behaviors are documented for the system team:

- **Clock stability:** the module requires a stable, continuous **131.072 kHz** `clk` with minimized slew on the distribution network to avoid glitch-induced FSM misbehavior.
- **POR handshake:** on release of the active-low asynchronous `rst_n`, the controller asserts `busy` and immediately begins stepping through OTP addresses 0–255.
- **OTP ROM interface timing:** the external OTP array must behave as a synchronous ROM – a one-cycle `otp_read_en` pulse at a given `otp_addr` must return stable data on `otp_data_out` exactly one `clk` edge later.
- **Output sampling:** all configuration outputs (`ro_trim_code`, both temperature-coefficient fields, calibration ratios, and control flags) hold at zero until the read sequence completes; downstream modules must gate their sampling on the rising edge of `done` rather than polling continuously.

Deliverables Checklist

The following packages were verified complete and committed for system integration, matching the Section 6.2 checklist of the project guideline:

- **RTL source files:** `otp_controller.v` (FSM, registers, patch-override logic), `crc16_ccitt.v` (CRC engine).
- **Testbench & simulation files:** `tb_otp_controller.v` (self-checking, all 6 DRS scenarios), `otp_sim_rom.v` (behavioral ROM model).
- **Synthesis scripts & constraints:** `syn_scr.tcl`, `lib_scr.tcl`, `otp_controller_clock_const.tcl` (131.072 kHz SDC).
- **Physical design hand-off databases:** `otp_controller_postsyn.v` (structural netlist), `otp_controller_postsyn.sdf` (back-annotation file), `otp_controller.invs_setup.tcl` (Innovus placement-and-routing setup).
- **Design reports:** area, timing, power, and gate-type profiling reports from Genus; RTL and GLS assertion logs.

Deliverables Status All items above are **[X] Complete**. Combined with the 39/39 RTL regression (Week 2), the 1,836-cell/51,831.965 μm^2 synthesized netlist meeting timing with 7.626 µs of slack (Week 3), and the 39/39 back-annotated GLS regression (Sec.), the OTP Controller module is **functionally verified, timing-closed, and packaged** for system integration.

Role Definitions & Task Allocation

To maximize efficiency during the final integration and reporting phase, responsibilities were divided among the group members according to their core technical strengths:

- Sina Hashemi **(Role A – RTL Design & Synthesis):** RTL debugging and FSM timing adjustments, and led the cross-group code-compatibility correction with Group 7 (Sec.).
- Mohammadhossein Sabzalian **(Role B – RTL Control & GLS Verification):** Compiled the structural verification data and analyzed the post-synthesis simulation checks for this report.
- Erik Hartouni **(Role C – Synthesis, Timing & Reporting):** Finalized the synthesis execution directory and placed the synthesis outputs in the deliverables package; co-authored the technical reports.
- Ali Behrad **(Role D – Testbench & Verification, Week 4 Integration Lead):** Led the final gate-level simulation run, extracted the raw simulation logs, collaborated with Role A on the Group 7 cross-verification, and compiled and submitted this final report.

Deliverables & Outlook

All Week 4 deliverables from the schedule were met: the gate-level netlist was re-validated under full SDF back-annotation (100% of path delays and width checks, 39/39 functional assertions passing, completion time matching the RTL reference to the picosecond); the FSM timing discrepancy surfaced by Group 7's cross-testbench exchange was corrected and re-verified against both test suites; system-integration notes were documented for the neighboring analog and frequency modules; and the full RTL, testbench, synthesis, and physical-design deliverables package was assembled and submitted by the Week 4 Integration Lead. With functional behavior, timing closure, and cross-group compatibility all confirmed, the OTP Controller (Group 8) is ready for hand-off to the system integration team.